PET ENGINEERING COLLEGE

DEPARTMENT-WISE ACADEMIC SYLLABUS

==

1. COMPUTER SCIENCE AND ENGINEERING (CSE)

==

Semester 1:

- Engineering Mathematics I
- Engineering Physics
- Engineering Chemistry
- Programming in C
- Engineering Graphics
- Professional English

Semester 2:

- Engineering Mathematics II
- Data Structures
- Object Oriented Programming
- Digital Principles and Computer Organization
- Environmental Science
- Technical English

Semester 3:

- Discrete Mathematics
- Database Management Systems
- Computer Networks
- Operating Systems
- Design and Analysis of Algorithms
- Object Oriented Software Engineering

Semester 4:

- Software Engineering
- Web Technology
- Artificial Intelligence
- Computer Architecture
- Theory of Computation
- Probability and Statistics

Semester 5:

- Machine Learning
- Cloud Computing
- Cyber Security
- Distributed Computing
- Mobile Application Development
- Professional Elective

Semester 6:

- Deep Learning
- Big Data Analytics
- Internet of Things
- Full Stack Development
- Professional Elective
- Mini Project

Semester 7:

- Project Work Phase I
- Technical Seminar
- Professional Elective
- Open Elective

Semester 8:

- Project Work Phase II
- Internship / Project Evaluation

==

2. ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (AIDS)

==

Semester 1:

- Engineering Mathematics I
- Engineering Physics
- Engineering Chemistry
- Programming in C
- Professional English
- Engineering Graphics

Semester 2:

- Engineering Mathematics II
- Python Programming
- Data Structures
- Digital Principles
- Technical English
- Programming Laboratory

Semester 3:

- Probability and Statistics
- Database Management Systems
- Object Oriented Programming
- Data Visualization
- Computer Networks
- Design and Analysis of Algorithms

Semester 4:

- Artificial Intelligence
- Machine Learning
- Operating Systems
- Web Technology
- Data Mining
- Software Engineering

Semester 5:

- Deep Learning
- Big Data Analytics
- Natural Language Processing
- Computer Vision
- Cloud Computing
- Professional Elective

Semester 6:

- Generative AI
- Reinforcement Learning
- MLOps
- Business Intelligence
- Mini Project
- Professional Elective

Semester 7:

- Project Work Phase I
- Technical Seminar
- Open Elective
- Professional Elective

Semester 8:

- Project Work Phase II
- Internship / Project Evaluation

==

3. MECHANICAL ENGINEERING (MECH)

==

Semester 1:

- Engineering Mathematics I
- Engineering Physics
- Engineering Chemistry
- Engineering Graphics
- Programming Fundamentals
- Professional English

Semester 2:

- Engineering Mathematics II

- Engineering Mechanics
- Materials Science
- Manufacturing Processes
- Technical English
- Engineering Drawing

Semester 3:

- Thermodynamics
- Fluid Mechanics
- Strength of Materials
- Manufacturing Technology
- Machine Drawing
- Mechanical Engineering Laboratory

Semester 4:

- Theory of Machines
- Heat Transfer
- Metrology and Measurements
- Manufacturing Technology II
- Engineering Materials
- Thermal Engineering Laboratory

Semester 5:

- Design of Machine Elements

- Dynamics of Machinery
- Automobile Engineering
- CAD/CAM
- Industrial Engineering
- Professional Elective

Semester 6:

- Finite Element Analysis
- Robotics and Automation
- Refrigeration and Air Conditioning
- Production Planning
- Mini Project
- Professional Elective

Semester 7:

- Project Work Phase I
- Technical Seminar
- Professional Elective
- Open Elective

Semester 8:

- Project Work Phase II
- Internship / Project Evaluation

===

4. CIVIL ENGINEERING (CIVIL)

   ===

Semester 1:

- Engineering Mathematics I
- Engineering Physics
- Engineering Chemistry
- Engineering Graphics
- Basic Engineering
- Professional English

Semester 2:

- Engineering Mathematics II
- Engineering Mechanics
- Surveying
- Building Materials
- Technical English
- Engineering Drawing

Semester 3:

- Strength of Materials
- Fluid Mechanics

- Structural Analysis
- Concrete Technology
- Geotechnical Engineering
- Surveying Laboratory

Semester 4:

- Design of Reinforced Concrete Structures
- Hydrology
- Environmental Engineering
- Transportation Engineering
- Soil Mechanics
- Concrete Laboratory

Semester 5:

- Structural Design
- Foundation Engineering
- Water Resources Engineering
- Highway Engineering
- Construction Management
- Professional Elective

Semester 6:

- Advanced Structural Engineering
- Environmental Management

- Quantity Surveying
- Remote Sensing and GIS
- Mini Project
- Professional Elective

Semester 7:

- Project Work Phase I
- Technical Seminar
- Open Elective
- Professional Elective

Semester 8:

- Project Work Phase II
- Internship / Project Evaluation

==

5. ELECTRONICS AND COMMUNICATION ENGINEERING (ECE)

   ==

Semester 1:

- Engineering Mathematics I
- Engineering Physics
- Engineering Chemistry

- Programming in C
- Engineering Graphics
- Professional English

Semester 2:

- Engineering Mathematics II
- Circuit Theory
- Electronic Devices
- Digital Electronics
- Signals and Systems
- Technical English

Semester 3:

- Analog Circuits
- Digital Signal Processing
- Electromagnetic Fields
- Microprocessors and Microcontrollers
- Communication Systems
- Electronic Circuits Laboratory

Semester 4:

- Control Systems
- VLSI Design
- Antennas and Wave Propagation

- Embedded Systems
- Computer Networks
- Communication Laboratory

Semester 5:

- Wireless Communication
- Digital Communication
- IoT
- Microwave Engineering
- Image Processing
- Professional Elective

Semester 6:

- Optical Communication
- Embedded System Design
- Machine Learning for ECE
- Robotics
- Mini Project
- Professional Elective

Semester 7:

- Project Work Phase I
- Technical Seminar
- Professional Elective

- Open Elective

Semester 8:

- Project Work Phase II
- Internship / Project Evaluation

==

6. ELECTRICAL AND ELECTRONICS ENGINEERING (EEE)

==

Semester 1:

- Engineering Mathematics I
- Engineering Physics
- Engineering Chemistry
- Programming Fundamentals
- Engineering Graphics
- Professional English

Semester 2:

- Engineering Mathematics II
- Circuit Theory
- Electrical Machines I
- Electronic Devices

- Digital Logic Circuits
- Technical English

Semester 3:

- Electrical Machines II
- Power Systems I
- Measurements and Instrumentation
- Power Electronics
- Control Systems
- Electrical Machines Laboratory

Semester 4:

- Power Systems II
- Electrical Drives
- Microprocessors
- Digital Signal Processing
- Renewable Energy Systems
- Power Electronics Laboratory

Semester 5:

- High Voltage Engineering
- Smart Grid
- Power System Protection
- Industrial Automation

- Electrical Vehicle Technology

- Professional Elective

Semester 6:

- Renewable Energy Technologies

- Power Quality

- Energy Management

- Embedded Systems

- Mini Project

- Professional Elective

Semester 7:

- Project Work Phase I

- Technical Seminar

- Professional Elective

- Open Elective

Semester 8:

- Project Work Phase II

- Internship / Project Evaluation

==

7. MASTER OF BUSINESS ADMINISTRATION (MBA)

==

Semester 1:

- Management Principles
- Managerial Economics
- Accounting for Managers
- Organizational Behaviour
- Business Statistics
- Business Communication

Semester 2:

- Marketing Management
- Financial Management
- Human Resource Management
- Operations Management
- Business Research Methods
- Management Information Systems

Semester 3:

- Strategic Management
- Entrepreneurship Development
- Project Management
- Elective I
- Elective II

- Summer Internship

Semester 4:

- Business Analytics
- International Business
- Elective III
- Elective IV
- Project Work
- Viva Voce

===

8. MASTER OF COMPUTER APPLICATIONS (MCA)

===

Semester 1:

- Programming in C
- Mathematical Foundations
- Computer Organization
- Database Management Systems
- Web Programming
- Programming Laboratory

Semester 2:

- Data Structures
- Object Oriented Programming
- Operating Systems
- Computer Networks
- Software Engineering
- Web Technology Laboratory

Semester 3:

- Advanced Java
- Artificial Intelligence
- Machine Learning
- Cloud Computing
- Data Analytics
- Professional Elective

Semester 4:

- Big Data Analytics
- Cyber Security
- Mobile Application Development
- Full Stack Development
- Professional Elective
- Mini Project

Semester 5:

- Project Work Phase I
- Technical Seminar
- Open Elective
- Professional Elective

Semester 6:

- Project Work Phase II
- Internship / Project Evaluation